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Bioprinting the future using light: A review on photocrosslinking reactions, photoreactive groups, and photoinitiators

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ABSTRACT

Light-based bioprinting is a type of additive manufacturing technologies that uses light to control the formation of biomaterials, tissues, and organs. It has the potential to revolutionize the adopted approach in tissue engineering and regenerative medicine by allowing the creation of functional tissues and organs with high precision and control. The main chemical components of light-based bioprinting are activated polymers and photoinitiators. The general photocrosslinking mechanisms of biomaterials are described, along with the selection of polymers, functional group modifications, and photoinitiators. For activated polymers, acrylate polymers are ubiquitous but are made of cytotoxic reagents. A milder option that exists is based on norbornyl groups which are biocompatible and can be used in self-polymerization or with thiol reagents for more precision. Polyethylene-glycol and gelatin activated with both methods can have high cell viability rates. Photoinitiators can be divided into types I and II. The best performances for type I photoinitiators are produced under ultraviolet light. Most alternatives for visible-light-driven photoinitiators were of type II, and changing the co-initiator along the main reagent can fine-tune the process. This field is still underexplored and a vast room for improvements still exist, which can open the way for cheaper complexes to be developed. The progress, advantages, and shortcomings of light-based bioprinting are highlighted in this review, with special emphasis on developments and future trends of activated polymers and photoinitiators.

1. Introduction

Additive manufacturing, also more commonly known as three-dimensional (3D) printing, is a collection of processes that have in common the use of successive additions of layers of the desired material to compose the final shape instead of the more traditional subtractive methods, in which portions of a blank are removed to sculpt the desired piece [1]. Those techniques make heavy use of computer-aided design to precisely control the bottom-up process, which means that most of the 3D printing methods were recently developed [2]. Due to its versatility, additive manufacturing allows the production of personalized and/or complex shapes, which is why it is widely used in the development of therapeutic approaches for personalized medicine [3]. For example, prosthetics can be made at the correct size and shape to fit the damaged tissue around an injury, or even to substitute members and internal organs with smaller risk of rejection, as they can be printed using bioinks embedding the patients' own cells. 3D printing can also be used to increase surgical success rates by producing patient-specific anatomic replicas that can be used for surgical planning and/or navigation [4,5].

Those replicas can be precisely created from imaging data obtained directly from the patient's anatomy and can allow a better view of the internal state of organs in a non-invasive procedure.

Among the many 3D printing technologies, bioprinting encompasses all printing processes using materials containing living cells or cell aggregates [6]. Bioprinting techniques require a polymeric framework in which the cells can be immobilized, without hindering their viability or their desired function. In addition to cells, nanoparticles and bioactive materials can also be embedded in the bioink to improve its physico-chemical and biological properties [7–10]. Although there are many alternatives for bioprinting nowadays, it is possible to classify them into four main groups [6–8]. First, inkjet-based bioprinting, which consists of liquid bioinks that can be extruded by heat or piezoelectric materials into a substrate whose contact will make it structurally rigid. Secondly, pressure-assisted microextrusion, for which the ink is already viscous enough to be directly deposited and cured in place after extrusion. Then, laser-assisted bioprinting, in which lasers are used to heat up the material to evaporate it prior to deposition. Finally, lithography-based bioprinting, such as stereolithography (SLA), whose main ad-

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vantages include fast printing speeds, high precision, and good surface finish.

The first 3D printing system was developed and reported by Dr. Hideo Kodama in 1981 [11]. However, because he missed the deadline to file the patent, it was rejected. Then in 1986, the first 3D printing patent was awarded to Chuck Hull for his SLA device [12]. The first step in SLA 3D printing is to create a 3D model of the (to be) printed construct using specialized 3D modeling software, or by using a 3D scanner to create a digital replica of an existing object. Once the 3D model is created, it is then sliced into a series of horizontal layers, which the 3D printer will use as a guide to build the object layer by layer using a photopolymer resin. This resin is photosensitive, so the printer uses a laser or other light source to cure and harden each layer of resin as it is deposited, ultimately creating a 3D physical object [13].

SLA technique can create high-resolution patterns by employing light to crosslink a photosensitive resin [14]. In this technique, the polymeric substrate is solidified by promoting light-induced crosslinks between its molecular chains, without the use of acid/base catalysts that are cytotoxic. The polymerization process is initiated by a photosensitive material called a photoinitiator which is initially added to the bioink. Free radicals are generated by the photoinitiators upon incidence of a specific frequency of light, that can locally promote crosslinking without making a chain reaction throughout the whole structure. This way, by precisely directing the crosslinking light, it is possible to precisely control the spatiotemporal solidification of the bioink [6].

Although, SLA bioprinting systems are of high speed, high resolution, high cell viability, and reproducibility, they usually rely on ultraviolet (UV) or near-UV light to offer enough energy to promote chemical reactions, which are harmful to cells [8]. So, creating novel visible-light-crosslinkable bioinks to replace UV-light-crosslinkable ones might solve these drawbacks. After defining the optimal wavelengths and curing time, the light incidence can be controlled by a series of digital micromirror devices, or DMDs, in a process invented by Larry J. Hornbeck in 1987, called Digital Light Processing (DLP) [6,15]. The position of those arrays of mirrors can be controlled by the applied voltage [16], allowing for the precise focusing of light and, with that, the high resolution bioprinting of the construct.

Two-photon polymerization (2PP) is another 3D lithography-based technique that uses a laser to cure photopolymer resins. It is similar to SLA in that it uses a laser to cure the resin, but it differs in that it uses two photons of light simultaneously, rather than a single photon, to polymerize the photoresin at the bulk of the liquid and not at the surface like SLA [17]. This allows for the creation of complex 3D structures with high resolution and fine features in the submicrometer range, but it requires a specialized setup and is generally slower than other 3D printing techniques.

A new light-based bioprinting technology, known as computed axial lithography or volumetric bioprinting), that can print high-resolution constructs within seconds has been developed recently [18]. This technology solves the layer-by-layer limitation of DLP and SLA techniques by rapidly bioprinting volumetric 3D functional constructs [19,20].

Here, we first review the mechanisms associated with crosslinking reactions taking place in materials used for light-based bioprinting. We then focus on the available options for activated polymer chains and photoinitiators and their influence on cell viability. Finally, the challenges and potential opportunities in the field of light-based bioprinting are also outlined.

2. Photocrosslinking reactions

Photocrosslinking reactions, also known as photopolymerization reactions, are chemical reactions that involve the use of light to create bonds between molecules and are often used in the synthesis of polymeric materials [21]. In a photocrosslinking reaction, a monomer or oligomer exposed to light is the building block used to form the poly-

mer. The light used in photocrosslinking reactions is typically in the UV or visible light spectrum.

Photocrosslinking allows the rapid and reversible modification of materials without the need for heat or other harsh chemical reactions. This makes it an attractive fabrication method for drug delivery, biosensors, and tissue engineering applications [22–24]. In addition, photocrosslinking can be used to create self-healing materials, which have the ability to repair damage caused by mechanical stress or exposure to harsh environments [25].

One of the main advantages of photocrosslinking reactions is that they can be fast, highly controlled, and precise, allowing for the creation of materials with specific properties and structures. However, materials used in this process can be expensive and mechanically weak. Overall, photocrosslinking reactions are an important tool in the synthesis and bioprinting of 3D polymeric materials from bioinks or bioresins precursors. These reactions can be sub-divided into: free-radical chain polymerization, thiol–ene photocrosslinking, and photomediated redox crosslinking.

2.1. Free-radical chain polymerization

Free-radical chain polymerization (Fig. 1A, D) is the common polymerization reaction used of photoreactive materials to form crosslinked hydrogels that typically proceeds through three stages: initiation, propagation and termination [26]. During the initiation stage, light breaks down the photoinitiator into free radical species in a process known as light-induced cleavage or photolysis. These free radicals will then react with monomers to effectively form new covalent bonds. In the propagation stage, free radicals react with additional monomer molecules, forming new radicals and continuing the polymerization process. Finally, in the termination stage, the radicals react with each other, terminating the polymerization reaction and forming the final crosslinked polymeric material.

The free-radical chain polymerization is a rapid reaction that allows fast crosslinking of polymeric materials. However, one potential issue with photocrosslinking is oxygen inhibition [27]. This occurs when oxygen molecules in the air come into contact with the photocrosslinking material before it has fully cured, which can lead to incomplete crosslinking. Incomplete crosslinking hinders shape fidelity post-bioprinting, which limit the successful printing of complex tissue structures [28]. Some chemical additives can play role in reducing oxygen inhibition, such as a higher concentration of photoinitiator, singlet oxygen generators and scavengers, gas generating radical initiators, hydrogen donors and other reducing agents, reactive methacrylates, hyperbranched polymers and dendrimers, and multifunctional acrylates [27].

2.2. Thiol–ene photocrosslinking

Thiol-ene photocrosslinking (Fig. 1B, E) is a type of chemical reaction in which thiol and alkene groups are exposed to light, causing them to react and form covalent bonds. Compared to free-radical chain polymerization, thiol-ene photocrosslinking allow an increased control over the crosslinking process, as the crosslinker concentration, length, and functionality can be altered to tune the mechanics, mesh size, and density of the crosslinked matrix. Moreover, oxygen attenuation of radicals is a major limitation for many photochemical reactions, but it can be solved using thiol–ene photoactivated chemistries [14]. Thiol–ene photoresins possess rapid reaction kinetics, which make them generate less harmful radicals while requiring a significantly lower polymer content, resulting in a better accommodating cellular matrix [29]. Furthermore, the native biopolymer bioactive properties are retained when using thiol-ene photocrosslinking, due to the significant reduction in the degree of substitution resulting from the step-growth crosslinking efficiency [29].

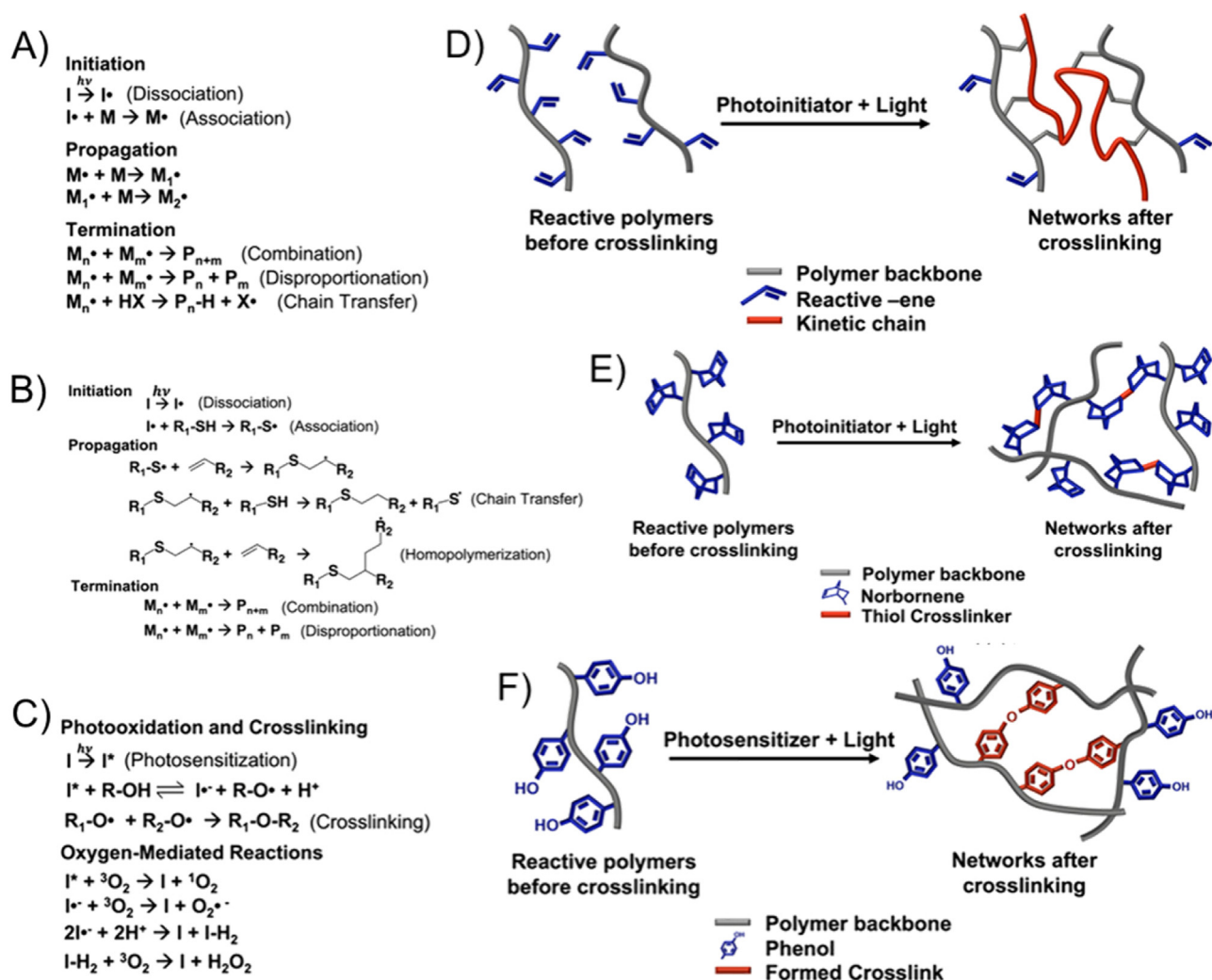


Fig. 1. General mechanism for (A) free-radical chain polymerization of bioinks and bioresins (B) radical-mediated thiol-ene photocrosslinking, and (C) photomediated redox reactions. Schematic of polymer chains containing reactive groups crosslinking through (D) free-radical chain polymerization, (E) radical-mediated thiol-ene reactions, and (F) photomediated redox reactions. Reproduced with permission from [26]. Copyright 2020, American Chemical Society.

It is important to note that mixed-mode photopolymerization, which is based on the concurrent reactions of thiol-ene reactions and free-radical chain polymerization, is possible by using functional groups associated with both polymerization mechanisms [26]. Acrylates or methacrylates can be crosslinked by the propagation of kinetic chains, while reacting with multifunctional thiol to also form crosslinks in a step growth manner. Rydholm *et al.* created thiol-acrylate degradable polymers of diacrylate poly(ethylene glycol) (PEG) monomer with thiol monomers resulting from a mixed-mode polymerization mechanism [30]. Precise control on the final network structure, properties, and degradation behavior of the material can be achieved by varying its functionalities.

2.3. Photomediated redox crosslinking

Photomediated redox crosslinking (Fig. 1C, F) of polymers modified with phenol groups takes place in the presence a photoredox catalyst used to initiate a reduction-oxidation reaction between reactive groups, causing them to form covalent bonds. Photosensitizers oxidize reactive groups after shifting into excited states following light absorbance [26]. Singlet oxygen and superoxides are produced when a photosensitizer absorbs a photon of light and becomes excited in aerobic conditions [31].

Although it is not a common reaction, hydrogen peroxide may as well be produced by photosensitizers, if two photons of light are absorbed in rapid succession. As the energy from both photons can be transferred to an oxygen molecule, resulting in the formation of hydrogen peroxide. Interestingly, it has been shown that photocrosslinking required the presence of oxygen, as oxygen is required to mediate the photooxidation and photocrosslinking of phenol groups [26,32]. This is one of the main differences between photomediated redox crosslinking and free-radical chain polymerization, as in the latter, photocrosslinking is inhibited when oxygen is present.

3. Photoreactive groups

Reactive polymers are polymers that contain chemical groups that can undergo a chemical reaction, such as crosslinking, to form a more complex polymer structure. Reactive polymers are obtained from monomers using a polymerization reaction, or through modification of existing polymers by the introducing of reactive groups onto their backbones [33]. This modification can be done through various chemical reactions, such as addition reactions, condensation reactions, or substitution reactions. Once the reactive groups are introduced, the polymer can then undergo crosslinking through a variety of methods, such as UV curing, thermal curing, or the use of a chemical crosslinking agent [34].

Most natural polymers need reactive groups to form crosslinked bonds, and those tend to be added in a pre-processing phase to react during the printing time.

3.1. Acrylate groups

Acrylates are a type of monomer that contain a double bond between the carbon atom of the acrylate group and a hydrogen atom [35]. This double bond is known as a "dangling double bond" because it is not bonded to any other atoms, making it highly reactive. As a result, acrylates can undergo a variety of chemical reactions, including polymerization [2]. Crosslinking of acrylate or acryloyl polymers using a photoinitiator involves exposing them to light, typically in the UV range, in the presence of a photoinitiator [36]. The photoinitiator absorbs the UV light and undergoes a chemical reaction that generates free radicals, which are highly reactive species that can initiate the crosslinking reaction (Fig. 2A). The free radicals react with the acrylate groups, forming chemical bonds between the different polymer chains and creating a three-dimensional network. Acryloyl or acrylate hydrogels are photocrosslinkable biomaterials that are widely used as bioprintable inks for tissue engineering applications [7,8]. Among those inks, the commercially available gelatin methacryloyl (GelMA) hydrogel is one of the best known for combining the biocompatibility of gelatin and the reactivity of the acryloyl group. GelMA is synthesized through the addition of methacrylate groups to the amine-containing side groups, then crosslinked in the presence of a photoinitiator under UV-light to form a hydrogel that is stable at 37 °C (Fig. 2C). Modified polyethylene glycol is also commercially available at a low price, but with less solubility and less compatibility with living tissues. Commercially available crosslinked inks have fewer issues with toxicity due to being purified from free reactive molecules before use, but the printing process still needs a minimum concentration of side chains available to generate crosslinked constructs. Many other acrylate derivatives exist. Additives can be used to create custom scaffolding environments and drug delivery systems with tunable properties [37–40]. Moreover, hybrid acrylated materials with biomimetic properties can be created to create bio-functional 3D constructs. For example, Nedunchezian *et al.* produced a novel photocurable hybrid hydrogel system comprising hyaluronic acid methacryloyl, GelMA, and acrylate-functionalized nano-silica crosslinkers for articular cartilage tissue engineering applications (Fig. 2D) [41]. They reported that their photocured hybrid hydrogel photocured hybrid hydrogel significantly increased chondrogenic marker gene expressions and enhanced type II collagen formation and the expression of sulfated glycosaminoglycan.

3.2. Norbornyl groups

If acrylate groups can undergo a free radical polymerization reaction when exposed to UV light in the presence of a photoinitiator, norbornyl groups can undergo a thiol-norbornyl crosslinking reaction when exposed to UV light in the presence of a thiol compound. Acrylates are normally considered cytotoxic, and there is less control over the overall structure [42]. For this reason, instead of using homopolymers, the thiol-ene reaction is under development. In the case of a thiol group and reactive carbon-carbon double bonds, the thiol group can act as a nucleophile, attacking the carbon-carbon double bond to form a new covalent bond [43]. This reaction typically requires the presence of a photoinitiator, which absorbs light and transfers the energy to the thiol group, causing it to become reactive (Fig. 2B). Alkenes, such as norbornene, acryloyl groups, vinyl sulfone, and maleimides are typically used for thiol-ene hydrogel formation [2]. Interestingly, by promoting the hydrolysis of the obtained gel, it is possible to recover the starting materials, making the thiol-norbornene-modified bioinks renewable [42].

4. Photoinitiators

In light-based bioprinting, photoinitiators are an essential component, as they are responsible for starting the polymerization process that allows the printer to create solid objects from liquid resin. Photoinitiators are typically added to the liquid resin pre-printing and are activated when the resin is exposed to light, which causes the resin to harden and solidify, and thus allowing the 3D printer to build up layers of material to create a 3D construct.

Overall printing time and cell viability are controlled by the choice of photoinitiator, as it affects the time and wavelength needed to photocrosslink the 3D printed scaffold. There are several different types of photoinitiators that are used in photopolymerization reactions. These include Type I photoinitiators, which absorb light in the ultraviolet (UV) range of the electromagnetic spectrum, and Type II photoinitiators, which absorb light in the visible range of the spectrum [2]. As can be seen in Table 1, many different parameters have been employed to photocrosslink light-based bioprinted constructs.

4.1. Type I photoinitiators—free radical precursors

When incident light is absorbed by type I photoinitiators, it can cause the molecules to reach an excited singlet or triplet state. This can lead to a process known as homolytic cleavage, in which the bonds within the molecules break, resulting in the formation of free radicals [26]. Cleavage usually occurs at the α -position of carbonyl groups, but can take place at any other weak bond. These free radicals can then go on to initiate the polymerization reaction. It is important to note that not all photoinitiators undergo homolytic cleavage when they are excited by incident light. For example, type II photoinitiators interact with a second component through an energy transfer or redox reaction to yield radicals [78]. Since type I photoinitiators consist of a single molecule, they are easier to control than type II photoinitiators. Cytocompatibility comparison of type I photoinitiators previously conducted by Bryant *et al.* concluded that 2-hydroxy-1-[4-(hydroxyethoxy)phenyl]-2-methyl-L-propanone (Irgacure 2959) lead to better chondrocytes viabilities than 2,2-dimethoxy-2-phenylacetophenone (Irgacure 651), 1-hydroxycyclohexyl phenyl ketone (Irgacure 184), and 2-methyl-1-[4-(methylthio)-phenyl]-2-(4-morpholinyl)-1-propanone (Irgacure 907), when encapsulated in PVA hydrogels [79]. In fact, Irgacure 2959 (Fig. 3A) was one of the first discovered type I photoinitiators and is still frequently used for bioprinting, with an activating frequency of 257 nm, well into the UV region [72]. As bioprinting inks must be compatible with the aqueous environment of the body, useful photoinitiators also need good solubility in water. Irgacure 2959 is of a hydrophilic nature and possess a low water solubility of 0.7% w/v. Although this concentration is low, it is usually enough to initiate the polymerization reaction.

Nonetheless, UV light (<370 nm) could lead to mutagenicity and phototoxicity [80–82]. For this reason, Irgacure 2959 has been mostly used with a light source of 365 nm to minimize those toxic effects [83]. However, this leads to a low reaction efficiency, as at 365 nm, Irgacure 2959 possess a very low molar extinction coefficient ($4 \text{ M}^{-1} \text{ cm}^{-1}$) [84]. Thus, to increase the polymerization efficiency, the exposure time and/or the light intensity should be increased, or a concentration of Irgacure 2959 should be used. Lithium phenyl-2,4,6-trimethylbenzoylphosphine (LAP) (Fig. 3B) is another type I photoinitiator that is commonly used as an alternative to Irgacure 2959. LAP has a better water solubility (<8.5% w/v) and a higher molar extinction coefficient at 365 nm ($218 \text{ M}^{-1} \text{ cm}^{-1}$) than Irgacure 2959 [84]. Thus, LAP absorbs a higher light amount at 365 nm, which leads to a higher initiation and polymerization rates rapid. Although increasing the concentration of either initiator increases its cytotoxicity, LAP maintains a good cell viability rate under much higher concentrations, which translates into shorter printing times [85]. Not

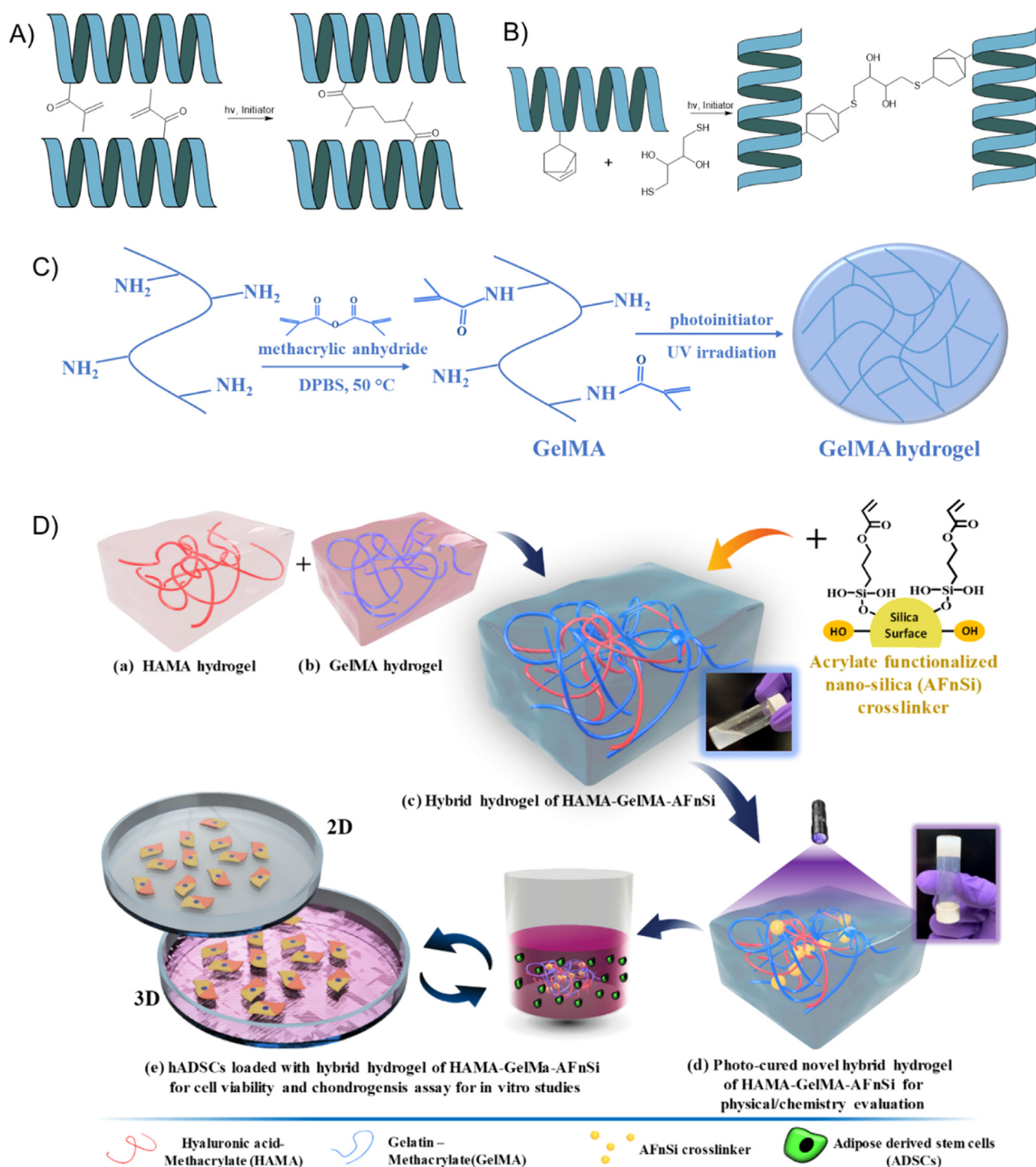


Fig. 2. Schematic representations of the (A) methacryloyl and (B) thiol-norbornyl crosslinking reaction, and (C) the preparation and crosslinking of gelatin methacryloyl (GelMA). (D) Schematic illustration of a novel photocurable hybrid hydrogel system comprising hyaluronic acid methacryloyl, GelMA, and acrylate-functionalized nano-silica crosslinkers used for articular cartilage tissue engineering applications. Reproduced from [41].

only is LAP less toxic than Irgacure 2959, but also the final product has better mechanical properties due to the presence of more crosslinks between the polymer chains. LAP also generates radicals under the incidence of 405 nm blue light, but its rate of reaction becomes much slower, on the order of days instead of minutes. This printing process, though, allows for high rates of living cells with small amounts of photoinitiator, which is a trade-off that can be considered [84].

4.2. Type II Photoinitiators – photosensitizers

The most promising photoinitiators that absorb visible light are, however, of type II, substances that are merely photosensitizers and get to an excited state under the absorption of light. Type II photoinitiators generate free radicals in the presence of a co-initiator through a multi-step reaction mechanism. In this process, the photoinitiator first absorbs light, which causes it to enter an excited state. The excited photoini-

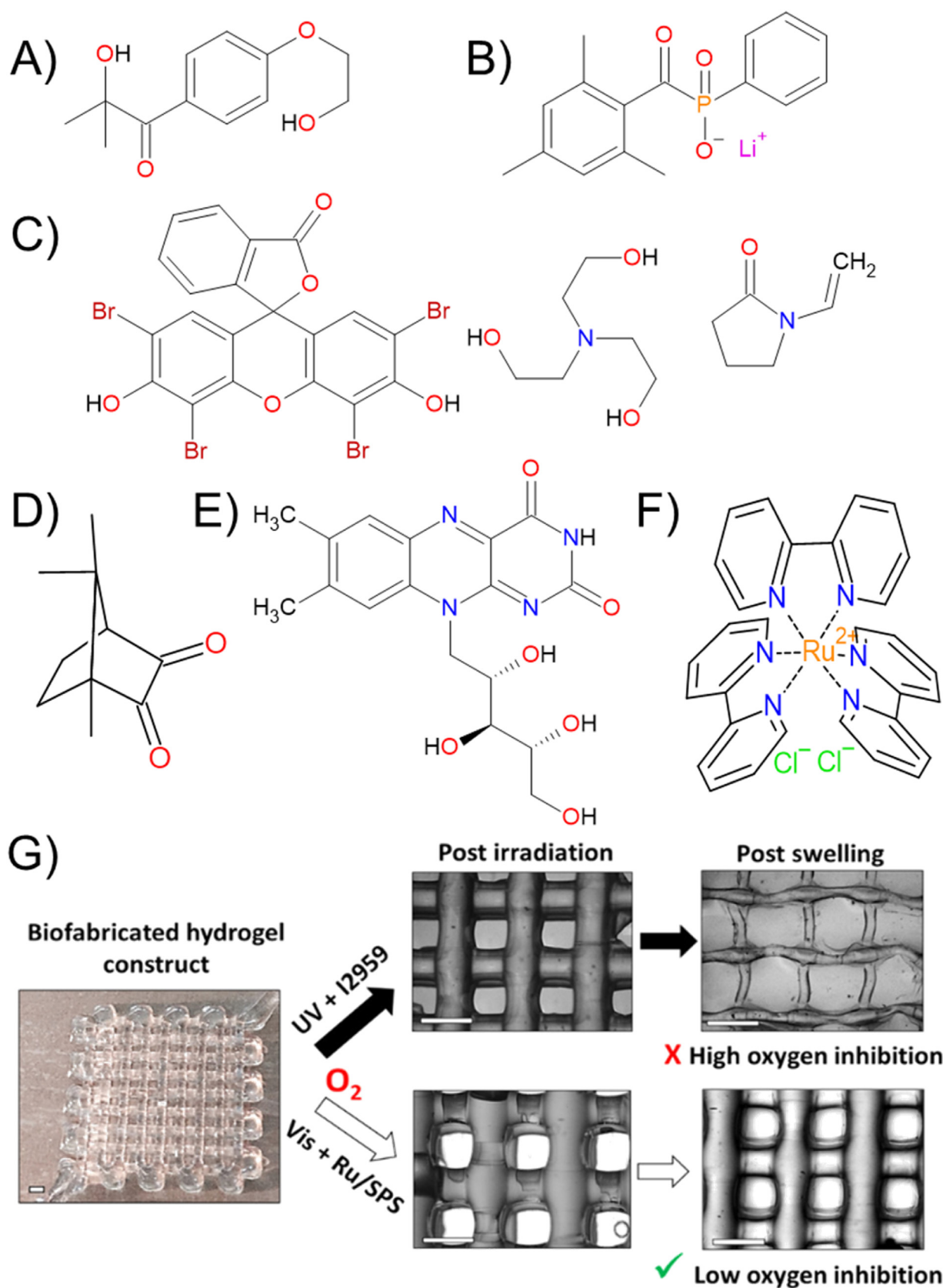


Fig. 3. The spatial arrangement of atoms in commonly used photoinitiators. A) Irgacure 2959. B) LAP. C) Eosin Y and its co-initiators TEOA and NVP. D) Camphorquinone, also known as 2,3-bornanedione. E) Riboflavin. F) Ru(bpy)₃Cl₂, the most common form of the complex. (G) Oxygen inhibition in 3D plotted 10 wt % GelMA and 0.6 wt % collagen hydrogels embedding Irgacure 2959 or ruthenium (Ru)/sodium persulfate (SPS) photoinitiators. Reproduced with permission from [28]. Copyright 2016, American Chemical Society.

Table 1
Photocrosslinking parameters used during light-based bioprinting using commonly employed photoinitiators.

Photoinitiator	Irgacure 2959	LAP	eosin-Y	Ru/SPS
Concentration	Range 0.01–1% w/v 0.05–1% w/v Most common 0.1% w/v 0.05% w/v	Range 0.05–0.6% w/v 0.037–0.6% w/v Most common 0.2% w/v 0.05% w/v	Range 0.01–10 mM Most common 0.01 mM	Range 0.2/2–1/10 mM
Intensity	Range 1.2–3950 mW/cm ²	Range 0.5–2000 mW/cm ²	Range 48–203 mW/cm ²	Range 3–50 mW/cm ²
Wavelength	Range 250–480 nm Most common 365 nm	Range 365–500 nm Most common 365 & 405 nm	Range 480–600 nm Most common ~500 nm	Range 400–450 nm
Duration	Range 2 s–30 min	Range 10 s–5 min	Range 2–20 min	Range 3–15 min
Ref.	[44–56]	[57–64]	[65–72]	[73–77]

tiator then transfers its energy to the co-initiator, which results in the formation of a radical species. In turn, those radicals attack the double bonds available in the polymer side chains, accomplishing the desired crosslink [6]. This technique, although slower, can accomplish >90% survivability frequently, and some of the used sensitizers are merely common pigments and food dyes (e.g. tartrazine, curcumin, and anthocyanin), which means that they are widely available for purchase, but their presence interferes with imaging processes [86,87].

Eosin Y (Fig. 3C), for example, is a water-soluble molecule that absorbs light in the 515 nm wavelength (green light) and is already used in microscopy for, among other uses, H&E stain, where it confers the cell cytoplasm a pink color to contrast with the blue-stained nucleus. This pigment can be used to cure a sample after being prepared without reducing cell viability, however, its radical generation is not fast enough to be used efficiently for light-based bioprinting [88,89]. There is evidence, though, that it is possible to use it for some light-based bioprinting techniques, such as SLA [72]. The best times for eosin Y are around minutes of curing, which is almost equivalent to the performance of UV-induced photoinitiators, with better cell survivability and better mechanical properties. Accelerators can be used alongside the co-initiator to generate more radicals and make a faster reaction, which means that with the correct concentrations of pigment and co-initiator, the reaction can become fast enough to be used as a viable 3D printing mechanism. Bahney *et al.* photoencapsulated human mesenchymal stem cells in PEGDA scaffolds using a visible light initiating system composed of the photosensitizer eosin Y, initiator triethanolamine (TEA), and catalyst 1-vinyl-2 pyrrolidinone (NVP) [90]. When compared to UV polymerization with Irgacure 2959, hydrogel scaffolds produced with this visible light photoinitiator showed a tighter crosslinked network three times faster and an increased viability of the encapsulated stem cells.

Another of the photoabsorbers commonly used to induce polymerization is camphorquinone, also known as 2,3-bornanedione (Fig. 3D). It is commonly used in resins for dental implants, implying some degree of biocompatibility [91]. However, experiments with living cells show a low viability rate of around 60%. Cell metabolism was even further reduced with the added co-initiators [79]. Although its absorbed wavelength of 470 nm is relatively safe compared to other photoinitiators, low cell viabilities and poor water solubility limit its use for 3D bioprinting.

Although the research on light-based bioprinting using visible light is still incipient, many alternatives for crosslinking parts printed in other techniques are known [6]. Most of the research on visible-light-induced light-based bioprinting was made using eosin Y, but many other options are available with good cell survivability indices. Riboflavin (Fig. 3E), also known as vitamin B2, is a biocompatible molecule that is used in many metabolic processes that involve redox reactions, commonly in

the forms of flavin mononucleotide (FMN) and the flavin adenine dinucleotide (FAD). The same ring structure that acts as a redox agent in the metabolism makes the molecule light-sensitive and ready to promote reactions, a property that can be used to generate free radicals similarly to eosin Y under 444 nm (blue light). Its cure time is also presented on the order of a few minutes [92], with improved mechanical properties using radiofrequency alongside the blue light or by increasing the exposure time. Although cell viability was not thoroughly studied, many studies showed this is a feasible technique, and the toxicity of this molecule is already known to be low due to its presence in the human body [93–95].

It is worth mentioning that there are a few other forms that type II photoinitiators can have. It seems that the most well-documented are metal complexes, which assume their well-known role as catalysts and can start cross-linking reactions under the presence of light. The most tested option was the system of tris(2,2-bipyridyl) dichlororuthenium(II) hexahydrate (Ru(bpy)₃Cl₂) (Fig. 3F) and sodium persulfate (Ru/SPS). The first one is activated under light to become a powerful oxidant, and the second regenerates the catalytic complex by releasing water. Overall, the cell viability with this method revealed itself as promising, with cell viability around 90% [96]. Lim *et al.* used wavelengths of 400–450 nm to irradiate Ru/SPS and to crosslink cell-laden constructs with high shape fidelity, cell viability, and metabolic activity [28]. They reported that the Ru/SPS system yielded better cell cytocompatibility than Irgacure 2959, even at high concentrations and visible-light irradiation intensities and for up to 21 days. In addition to the better cell cytocompatibility, a reduced oxygen inhibition was observed with increasing Ru/SPS photoinitiator concentration and light irradiation intensity, which is known to directly impact the 3D printing fidelity of photopolymerized hydrogels (Fig. 3G). Furthermore, the curing time of ruthenium is comparable to that of type I photoinitiators [28,96,97]. Notice that ruthenium, as a rare metal, is expensive, which renders the overall method promising but not economically viable with this specific compound.

5. Conclusions and future directions

In addition to biorthogonal photocrosslinking chemistries, this work focused on advancements regarding two other optimization points for light-based bioprinting: reactive side chain groups, added to the polymers for cross-linking, and light-sensitive photoinitiators, used to generate free radicals and initiate the reaction of side chain groups. The most common polymers used in the analyzed reports were polyethylene glycol and gelatin, with the former being easily synthesizable, and the latter completely biocompatible. To promote the crosslinking reaction, those polymers were treated with reagents that add electron-rich side

chain groups, such as acryloyl and norbornyl. The first one is more common but requires purification before being used as acrylates are toxic to cells. For this reason, they are normally offered as commercial products (PEGMA and GelMA) which are suitable for direct use. Both polymers can be directly crosslinked or, to allow for more precise control, polymerized with a thiol-containing reagent.

For photoinitiators, the fastest available options are still of type I, which are single molecules commonly cleaved with UV light. The most used initiator is Irgacure 2959, which allows for fast printing with viability >75% in most conditions, but is toxic in high concentrations. The best substitute is LAP, which was more recently developed and requires a weaker light wavelength, allowing for >90% survivability under the same conditions. Type II photoinitiators are still incipient in their use in light-based bioprinting, but they commonly allow >90% survivability due to the non-ionizing light needed for crosslinking. The most explored type II photoinitiators is eosin Y, which still requires minutes of curing for proper use, but commonly allows >95% survivability. Camphorquinone and Ru(bpy)₃²⁺ are two options proven effective, but they do not seem to be viable enough to replace Irgacure. Camphorquinone use result in low cell viabilities. Whereas, riboflavin is known for its biocompatibility, but has seldom been used so far in light-based bioprinting, and its curing time is exceedingly slow when used.

Much of the work on light-based bioprinting is recently dated, due to the computational requirements to make it feasible, which means that there is still large room for development. As for reactive groups, other options exist but remain largely unexplored. Most of the work is concentrated on functionalizing new polymers with acrylate rather than trying new ways to polymerize. Therefore, the research of new, safer crosslinking reagents is still largely unexplored and a possible research area, although smaller than that of polymer selection. However, the most overlooked improving point is regarding photoinitiators. Riboflavin showed itself a biocompatible but slow photosensitizer, and therefore it can be used to print with more sensitive cells in a variety of conditions without threatening their survivability. Metal complexes are also a promising area, for which biocompatibility and economic viability are the main issues to be resolved, as the kinetics of the reaction are comparable to the ones using UV-light.

Although it is difficult to predict the future with certainty, but based on current research and developments in the field, it is likely that light-based bioprinting will continue to play an important role in the field of tissue engineering and regenerative medicine. Volumetric bioprinting is a relatively new technology that has already solved the speed and layer-by-layer limitations of existing light-based bioprinting techniques. However, this technique along with other advanced bioprinting techniques, such as multi-material bioprinting and 4D bioprinting, still lack proper commercialization. In the future, we will likely see developments and commercialization of more advanced bioprinting techniques that use light to create highly detailed and vascularized volumetric structures to solve challenges in nutrient transport and metabolite exchange in bioprinted tissues and organs. Overall, the future of light-based bioprinting is expected to see significant advancements in terms of precision, accuracy, and complexity of the printed structures. With that, the transplantation of bioprinted functional organs, which was once considered science fiction, could become a reality.

Declaration of Competing Interest

The authors declare no conflict of interest.

CRediT authorship contribution statement

Kamil Elkhoury: Conceptualization, Investigation, Methodology, Visualization, Writing – original draft. **Julio Zuazola:** Conceptualization, Investigation, Methodology, Visualization, Writing – original draft. **Sanjairaj Vijayavenkataraman:** Conceptualization, Supervision, Project administration, Writing – review & editing.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:[10.1016/j.slas.2023.02.003](https://doi.org/10.1016/j.slas.2023.02.003).

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